

UNIVERSITY OF ZIMBABWE

Department of Mathematics and Computational Sciences

MSc Actuarial Sciences

MSAC512 MACHINE LEARNING

PRACTICAL ASSIGNMENT 1

**Predictive Modelling of Residential House Prices in Zimbabwe Using Web-Scraped Structured,
Textual and Visual Data**

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Programme: MSc Actuarial Sciences

Module: MSAC512 Machine Learning

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Executive Summary

This report develops a multimodal automated valuation model (AVM) for residential houses advertised on property.co.zw. The notebook supplied for the assignment contains 524 scraped listings, of which 376 USD-denominated observations fall within the defined Greater Harare and commuter-town study area. The study covers 96 suburbs and explicitly includes Madokero and Mabvazuva. Structured attributes, free-text descriptions and property photographs are combined to explain and predict advertised house prices.

Text-derived condition and finish indicators were constructed using controlled regular-expression rules, while TF-IDF features captured broader description language. For the visual component, 150 listings with usable photographs were sampled across seven areas and processed with CLIP ViT-B/32 to obtain eight continuous condition/finish scores. The final modelling cohort therefore satisfies the requirement for a substantial multimodal image sample.

The formal suburb test shows a strong association between suburb and log advertised price after controlling for floor size and bedrooms ($F = 17.13$, $p < 0.001$; partial $\eta^2 = 0.781$). On the validation set, the Random Forest produced the lowest RMSE and MAE among the three machine-learning families evaluated, while Elastic Net achieved the highest log-scale R^2 . The final reconstructed Random Forest test evaluation reported RMSE of \$298,776.73, MAE of \$166,452.74, MAPE of 61.40%, log-RMSE of 0.6278 and log- R^2 of 0.6164.

For the applied case study, the model estimated \$95,349 for House A in Madokero and \$94,706 for House B in Mabvazuva. The corresponding 90% conformal intervals are very wide, reflecting material uncertainty. Density diagnostics place House A in a relatively common region of the training space and House B in a moderately sparse region. The results are therefore suitable as indicative model outputs rather than stand-alone professional valuations.

1. Introduction and Objective

Zimbabwean residential property valuation is challenging because asking prices reflect location, property size, title status, infrastructure, finish quality and marketing language. Conventional tabular models may omit useful information contained in listing descriptions and photographs. The objective of this assignment is therefore to build and evaluate a reproducible multimodal house-price model using structured, textual and visual information scraped from a Zimbabwean property portal.

The analysis follows the assignment structure: Section A covers data acquisition, cleaning and ethics; Section B covers multimodal feature engineering; Section C presents exploratory and statistical diagnostics; Section D develops an actuarial benchmark and three machine-learning model families; Section E evaluates interpretation, residual behaviour and fairness; and Section F applies the model to two partially finished houses in Madokero and Mabvazuva.

2. Section A — Data Acquisition, Pre-processing and Scraping Ethics

2.1 Data coverage

The raw notebook dataset contains 524 records from property.co.zw. The analysis then restricts attention to USD-denominated residential house listings in Greater Harare and selected commuter towns. The resulting analysis dataset contains 376 listings, 96 suburbs and nine study areas. Madokero contributes 2 observations and Mabvazuva contributes 7 observations.

Area	Listings	Median advertised price (USD)
Harare North	153	540000
Harare West	95	170000
Harare South	50	172500
Harare East	29	165000
Chitungwiza	16	38000
Norton	16	105000
Ruwa	15	115000
Damofalls	1	25000
Seke	1	15000

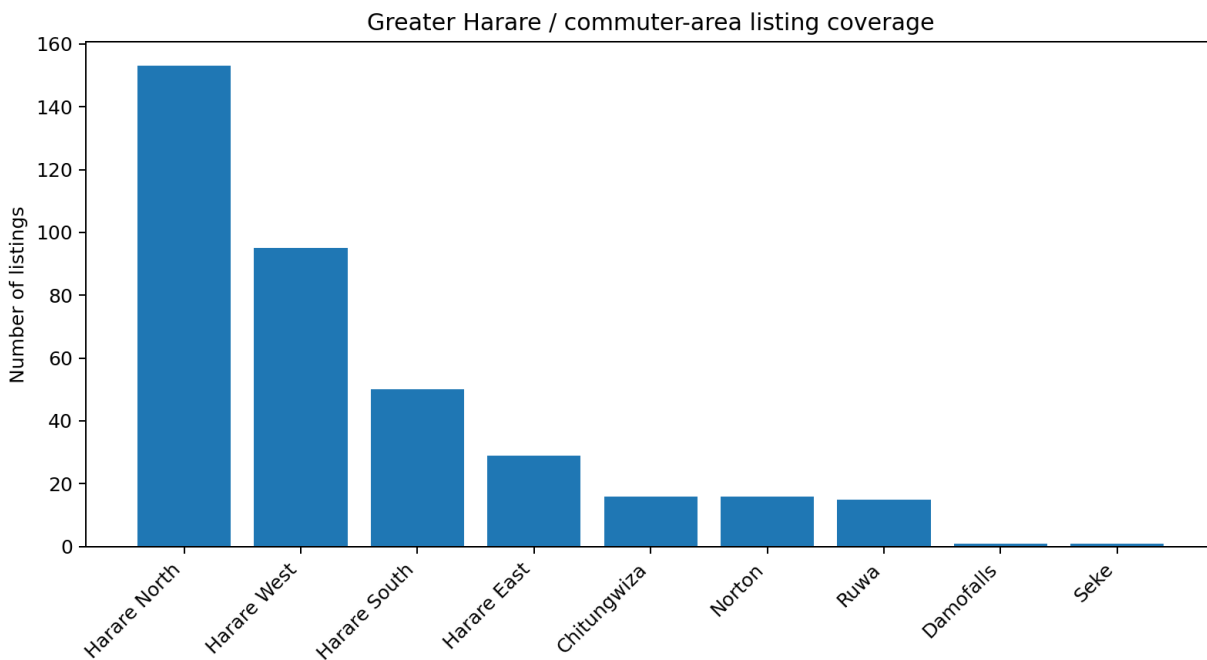


Figure 1. Listing coverage across Greater Harare and commuter areas.

2.2 Cleaning and quality controls

The notebook checks missingness, duplicate identifiers and duplicate URLs, then inspects implausible size values. Floor or stand sizes at or below 10 m² are treated as likely data-entry or unit problems and are set to missing for modelling rather than being accepted literally. Three listings have missing advertised prices and are retained in the raw data but excluded from price modelling. Duplicate checks are performed before the final analytical dataset is created.

The analysis deliberately restricts the target to USD-denominated listings rather than converting ZiG prices. This avoids introducing exchange-rate measurement error into the target and still leaves more than the required 300 observations. The consequence is that the fitted model represents the USD-advertised segment of the market rather than every Zimbabwean residential listing.

2.3 Scraping ethics and professional conduct

Data Collection Date: The property listings were collected from property.co.zw on 24 September 2026. The scraping process incorporated rate limiting, with a randomized delay of approximately 1–3 seconds between requests, to reduce server load. The website’s robots.txt file and terms of use were reviewed before data collection, and restricted or administrative paths were avoided. The data were collected strictly for academic purposes and were not intended for commercial resale or republication.

3. Section B — Multimodal Feature Engineering

3.1 Hand-built text indicators

A controlled vocabulary was used to convert description phrases into binary condition and finish indicators. The final dictionary includes paint/unpainted, paving, walling/security, ceiling/no-ceiling, fitted/unfitted kitchen, borehole, water tank, solar, ensuite, incomplete, tiled/bare floor, built-in cupboards, title deeds and cession. The approach is transparent and auditable, although sparse mention counts mean that absence of a flag should be interpreted as 'not mentioned' rather than proof that the feature is absent.

Observed phrase from listings	Mapped feature
incomplete 4 bedroomed house	incomplete_flag
prolific borehole	borehole_flag
6kva solar, borehole	solar_flag
fitted kitchen	kitchen_fitted_flag
main ensuite	ensuite_flag
walled and gated	walling_security_flag
tiled throughout	tiled_floor_flag
clean Title Deeds	title_deeds_flag
offered on cession	cession_flag
paved driveways	paving_flag
BICs	built_in_flag
ceilings, plastering and skimming done	ceiling_flag

3.2 TF–IDF textual representation

The notebook uses TfidfVectorizer with English stop words, 1–2 grams, min_df = 3, max_df = 0.90 and sublinear term frequency. In the final model pipeline the training vocabulary contains 182 terms and is reduced to 30 SVD components, explaining approximately 67.1% of the TF–IDF variance. A preliminary comparison reported log-RMSE values of 0.5940 for structured features only, 0.5713 after adding hand-built indicators, and 0.5692 after adding TF–IDF, suggesting that both engineered flags and broader language contribute incremental information.

3.3 Visual feature extraction

Of the 376 Greater Harare observations, 365 have at least one usable image, 362 have at least three, 349 have at least five, 286 have at least ten and 183 have at least fifteen. A stratified 150-listing sample was selected, including Madokero and Mabvazuva. The sample contains 62 Harare North, 39 Harare West, 19 Harare South, 11 Harare East, 7 Ruwa, 6 Norton and 6 Chitungwiza listings.

Minimum usable images	Listings
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1	365
3	362
5	349
10	286
15	183

CLIP ViT-B/32 was used to score photographs against paired prompts representing painted walls, paving, fitted kitchens, ceilings, finished floors, security walls/gates, solar systems and overall completion. Up to three images per listing were selected from the first, middle and last image positions. The resulting eight continuous visual scores were merged with the structured and text features.

Visual feature	Mean score
Paint	0.3290
Paving	0.2440
Kitchen	0.6690
Ceiling	0.2010
Floor	0.3890
Security	0.2660
Solar	0.2460
Completion	0.6780

4. Section C — Exploratory Data Analysis and Statistical Diagnostics

4.1 Price distribution and transformation

Across the 376-listing Greater Harare dataset, the mean advertised price is approximately \$362,952, the median is \$240,000 and the maximum is \$3,000,000. The substantial right tail makes direct-scale residuals highly heteroskedastic and motivates modelling $\log(\text{price})$. The log transformation compresses extreme high-value listings while retaining interpretable multiplicative effects.

Median floor size is approximately 350 m² and the median advertised price per square metre is approximately \$600. The median price per bedroom is approximately \$61,250.

4.2 Suburb-level price diagnostics

Suburb-level price per square metre varies materially. Selected medians are shown below. Madokero is retained descriptively despite having only two observations, while Mabvazuva has seven observations in the Greater Harare dataset.

Suburb	Median advertised price/m ² (USD)
Alexandra Park	1933
Borrowdale	1742
Borrowdale Brooke	1600
Mount Pleasant	1390
Quinnington	1358
Glen Lorne	1313
Avondale	1260
The Grange	1235
Mabvazuva	650
Madokero	573

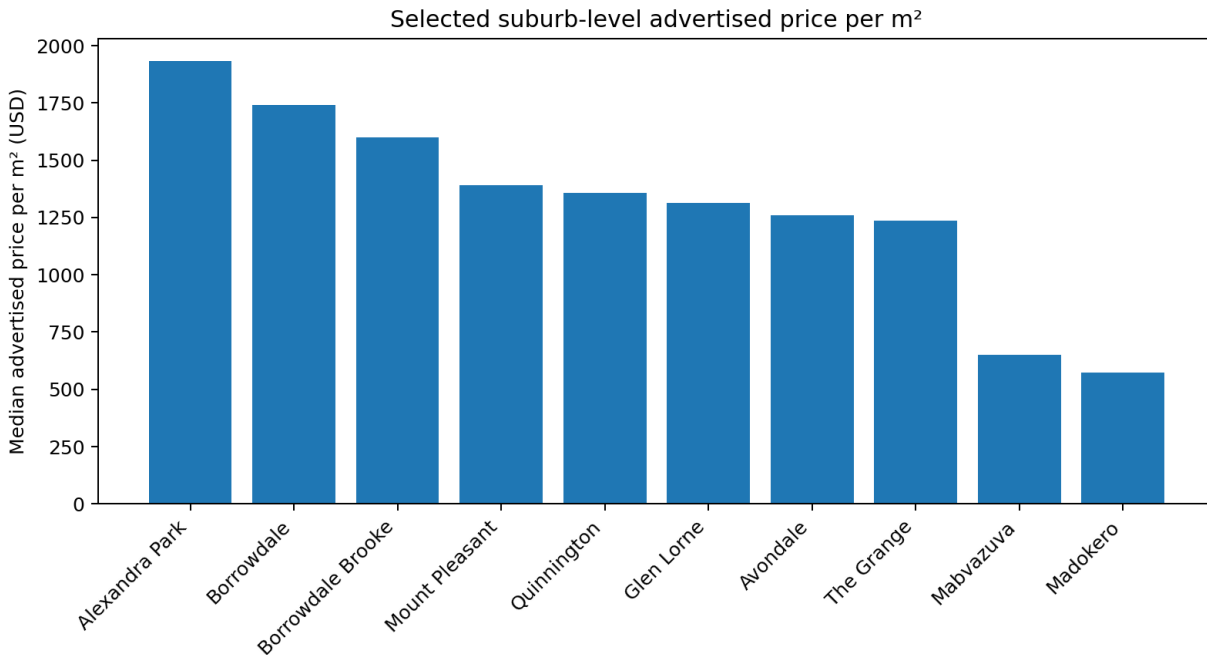


Figure 2. Selected suburb median advertised price per square metre.

4.3 Formal suburb association

Statistic	Value
F-statistic	17.13
p-value	0.0000
Partial eta-squared	0.7807

The OLS/ANOVA specification models log price using log floor size, bedrooms and suburb fixed effects. The overall suburb term is highly significant ($F = 17.13$, $p \approx 2.78 \times 10^{-53}$). The partial η^2 of 0.781 indicates a large share of explainable variation is associated with suburb after controlling for the included size variables. This is an association, not a causal effect.

4.4 Multicollinearity and condition-price patterns

Condition indicators with non-zero variance have VIFs of roughly 1.01–1.77, so the notebook finds no material multicollinearity problem among the retained binary flags. Zero-variance indicators are excluded from VIF calculations. The condition-price plot below demonstrates that the relationship between finish mentions and advertised price varies substantially across suburbs and is confounded by location and property type.

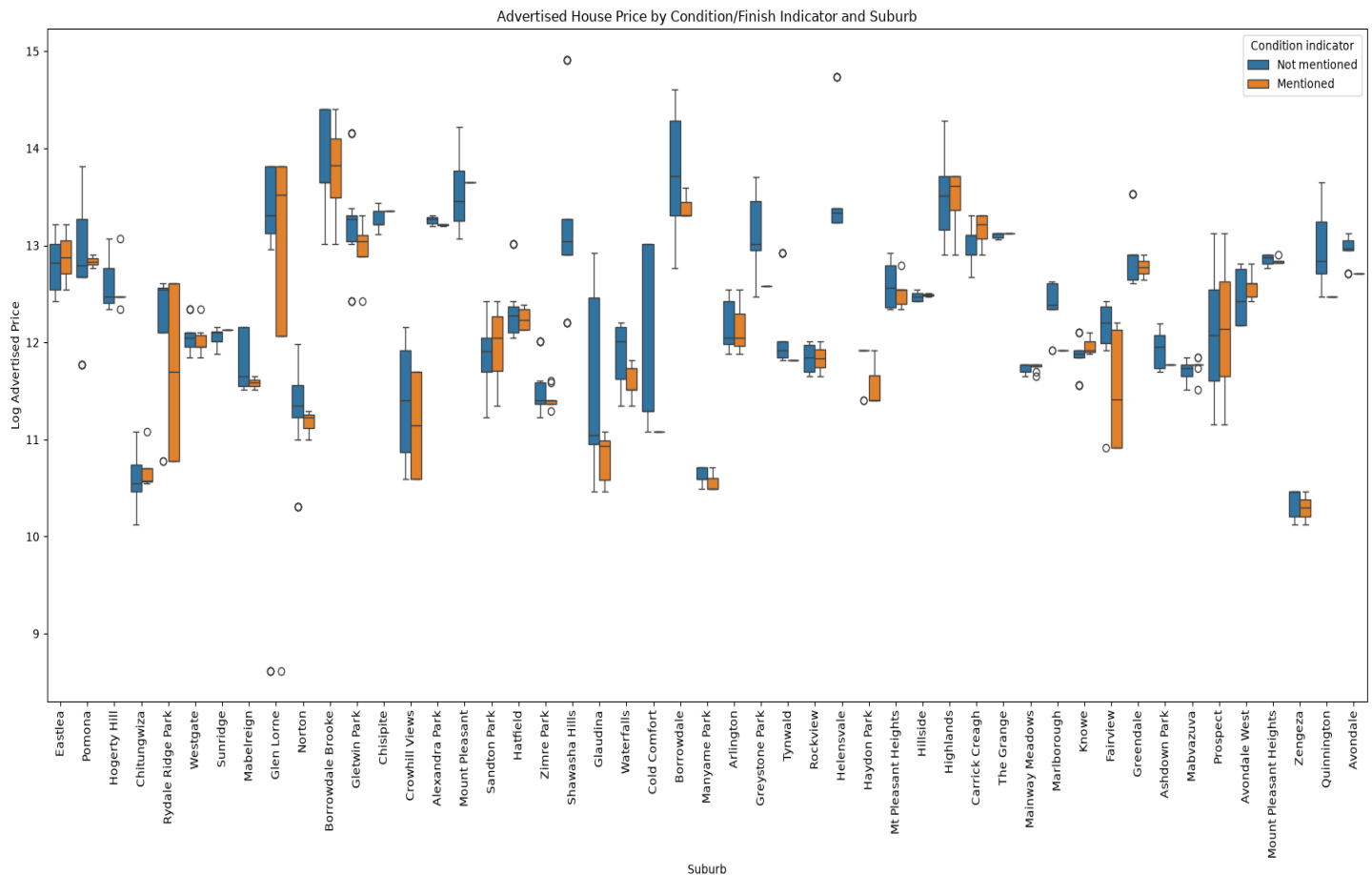


Figure 3. Log advertised house price by condition/finish mention and suburb (notebook output).

5. Section D — Predictive Model Development

5.1 Spatial split and leakage control

The multimodal modelling cohort contains 150 listings from 66 suburbs. A suburb-disjoint split is used: 96 training observations, 22 validation observations and 32 test observations. The notebook explicitly checks that there is no suburb overlap between training, validation and test sets. This is stricter than a naive random split and tests transfer to unseen suburbs.

Partition	Listings	Suburbs
Training	96	41
Validation	22	11
Test	32	14

5.2 Actuarial benchmark

The benchmark is a structured-feature model using location, bedrooms, bathrooms and property size. On the validation set it records RMSE \$999,765, MAE \$410,763, MAPE 110.86%, log-RMSE 0.8188, log-MAE 0.5905 and log-R² 0.3702. The relatively weak benchmark establishes the value of nonlinear and multimodal modelling, but the very high original-scale RMSE is also influenced by the small validation set and extreme upper-tail prices.

5.3 Machine-learning models and validation comparison

Model	RMSE (USD)	MAE (USD)	MAPE (%)	Log R ²
Actuarial benchmark	999765	410763	110.86	0.3702
Elastic Net	235018	165360	74.29	0.5732
Random Forest	214680	128540	80.43	0.5290
Gradient Boosting	238310	160779	96.65	0.4407

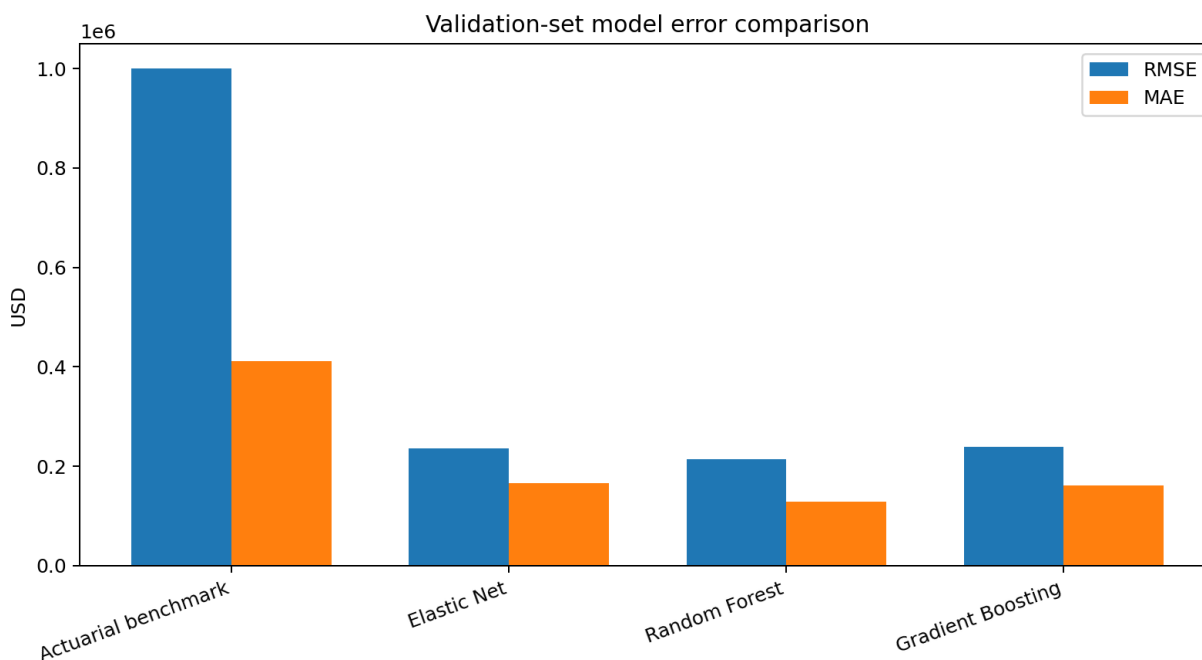


Figure 4. Validation-set RMSE and MAE across the benchmark and three ML families.

Three materially different machine-learning families are compared: Elastic Net, Random Forest and Gradient Boosting. Random Forest has the lowest validation RMSE and MAE. Elastic Net has the lowest MAPE and the highest log-R² among the three ML models. The notebook therefore selects Random Forest on the basis of its stronger absolute-error performance and acceptable uncertainty width rather than a single metric alone.

Model	Mean prediction-interval width (USD)
Elastic Net	1692307
Random Forest	1821347
Gradient Boosting	2130300

5.4 Final Random Forest test performance

Metric	Test result
RMSE	\$298,776.73
MAE	\$166,452.74
MAPE	61.40%
Log RMSE	0.6278
Log MAE	0.4776

Log R ²	0.6164
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The final reconstructed Random Forest uses 500 trees, max_depth = 5, max_features = 0.7 and min_samples_leaf = 1. The feature matrix contains structured variables, retained condition flags, eight visual scores and 30 SVD text components. The test metrics above are the final held-out results used in Sections E and F.

6. Section E — Evaluation, Interpretability and Fairness

6.1 Residual diagnostics

The residual-versus-fitted plot shows that errors become larger at the upper end of predicted prices, with a small number of very large positive residuals. This indicates systematic underprediction of some expensive properties. The floor-size diagnostic likewise shows that the largest residuals occur among some larger houses, although most observations cluster below roughly 600 m².

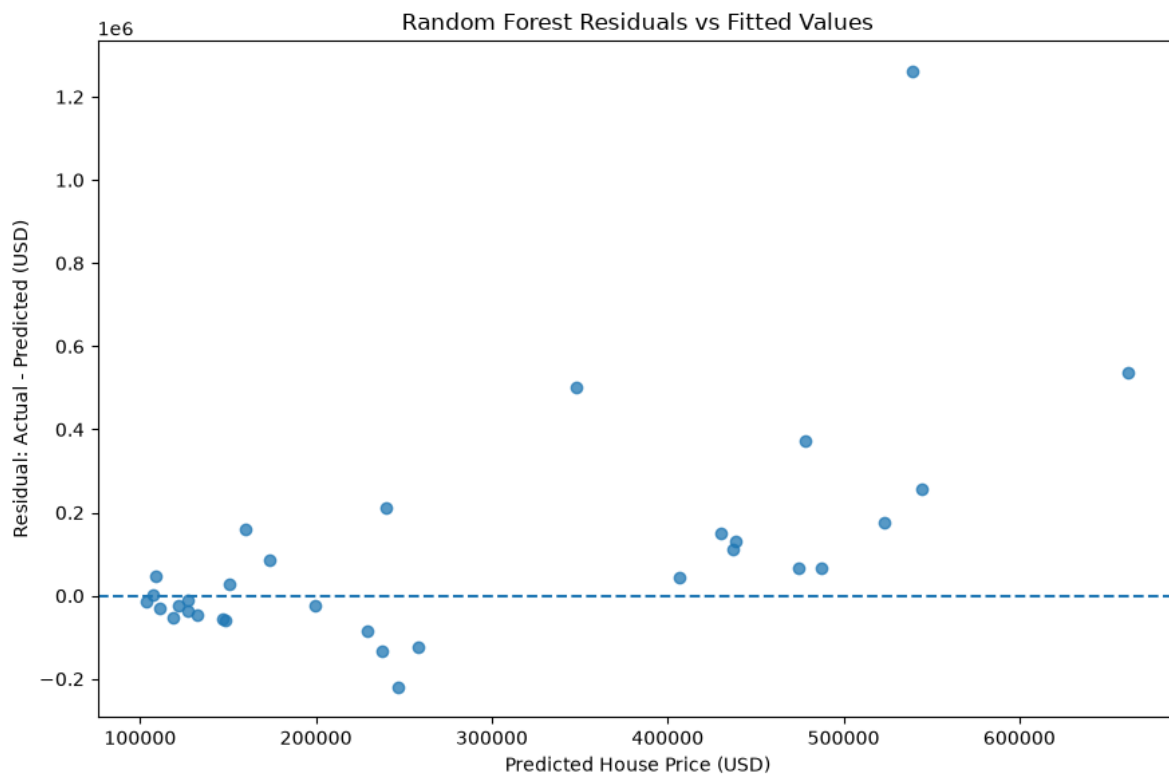


Figure 5. Final Random Forest residuals versus fitted values.

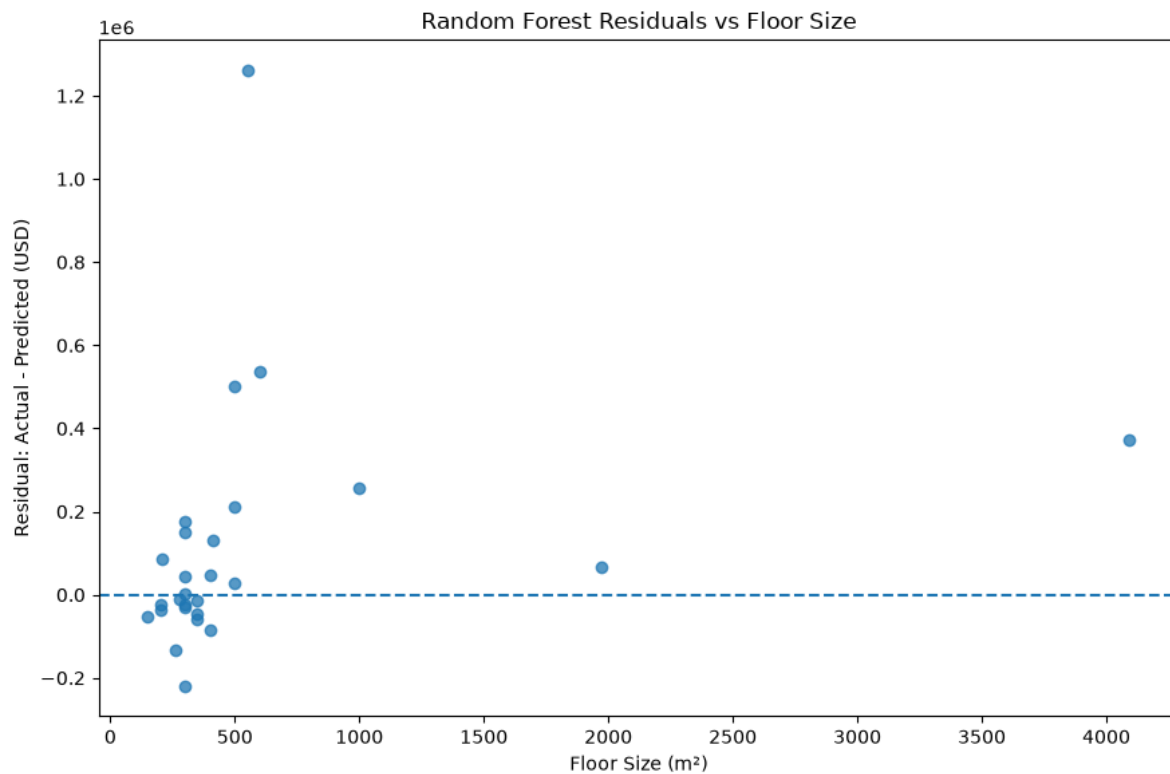


Figure 6. Final Random Forest residuals versus floor size.

6.2 Global importance

Feature	Permutation importance
area_Harare North	0.1885
stand_size_m2	0.1562
floor_size_m2	0.0166
bathrooms	0.0151
visual_completion_score	0.0131
visual_paving_score	0.0110
tfidf_svd_7	0.0042
tfidf_svd_14	0.0041
visual_paint_score	0.0033
bedrooms	0.0033

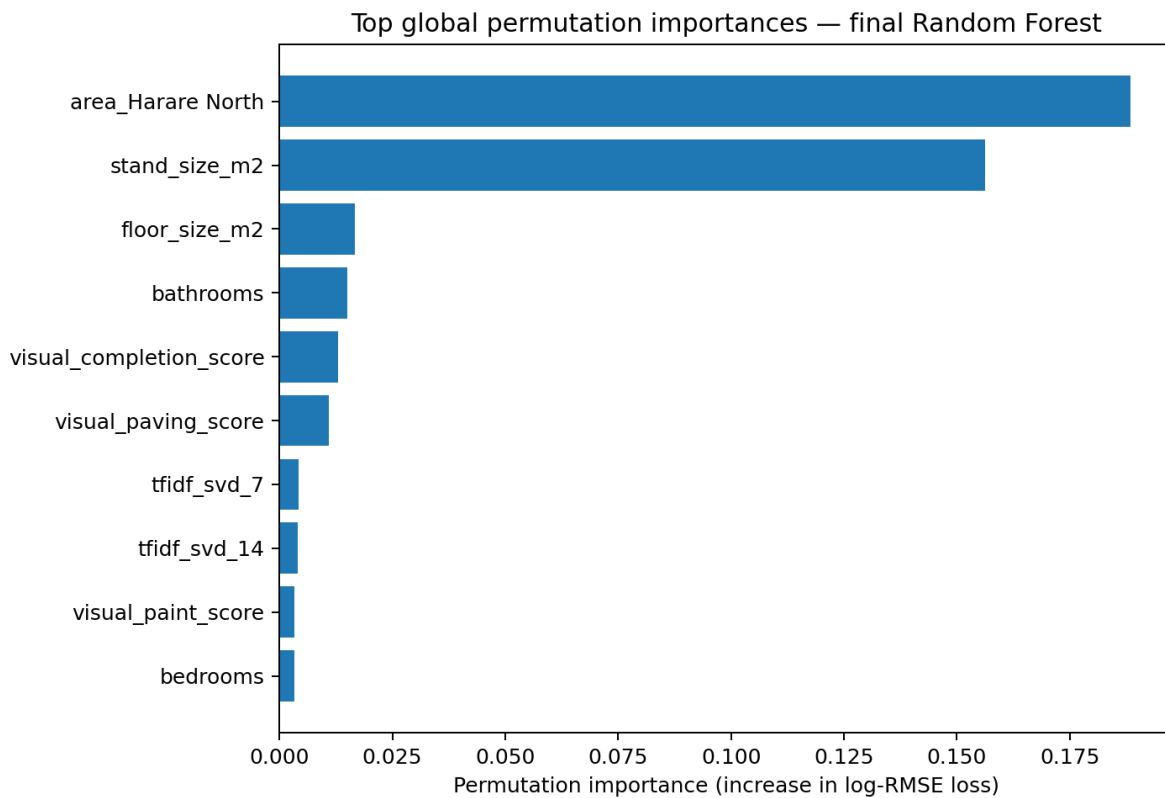


Figure 7. Top ten global permutation importances.

The strongest individual predictors are Harare North area membership and stand size. Floor size and bathrooms contribute smaller but still positive importance, while visual completion, paving and paint scores appear among the top predictors. Several SVD text components also contribute. These values measure predictive dependence in this fitted model and should not be interpreted causally.

6.3 Group contribution analysis

Model	RMSE (USD)	MAE (USD)	MAPE (%)	Log RMSE
Full model	298,776.73	166,452.74	61.40	0.6278
Without location	321,882.96	194,741.73	60.26	0.6541
Without size/bedrooms	308,570.51	170,449.02	72.35	0.6790
Without condition/finish	284,803.33	165,571.89	67.48	0.6338
Without text	319,775.14	186,232.28	64.17	0.6673

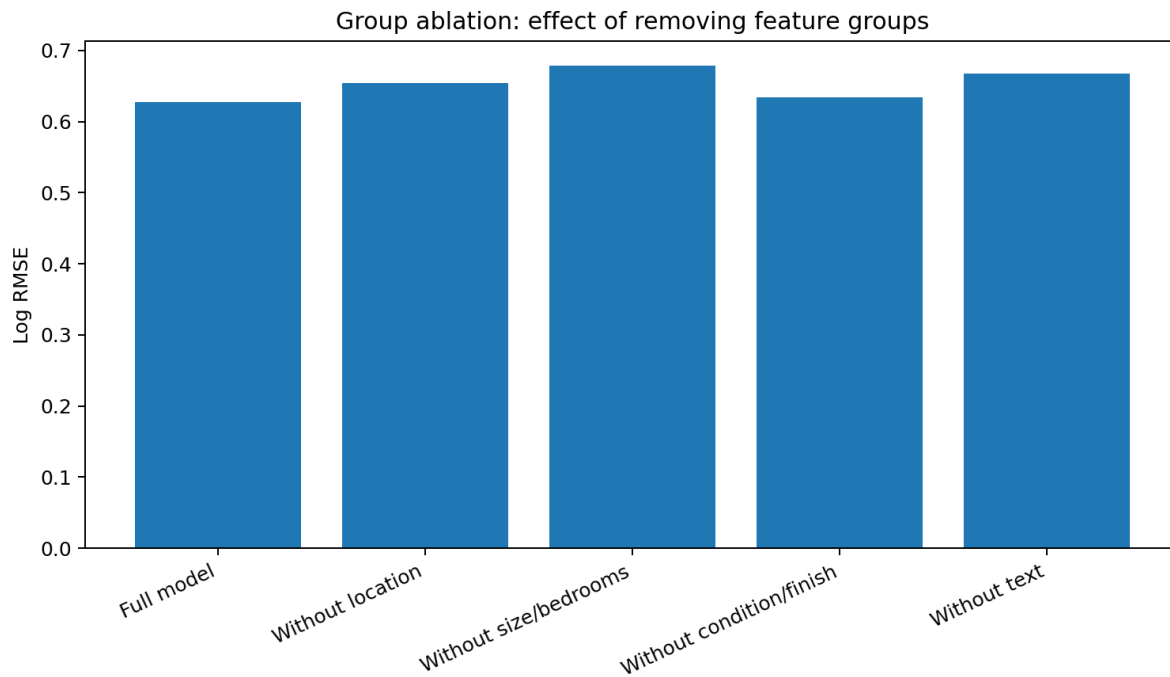


Figure 8. Group ablation measured by log-RMSE.

Using log-RMSE as the primary scale, removing size/bedrooms causes the largest deterioration (+0.0512), followed by text (+0.0395) and location (+0.0263). Removing condition/finish changes log-RMSE by only +0.0060. The raw RMSE actually improves when condition/finish is removed, illustrating that conclusions depend on the error scale and that sparse text flags may add noise for high-value observations.

6.4 Local explanations

Case	Identifier	Suburb	Actual (USD)	Predicted (USD)	Absolute error (USD)
Largest absolute error	KNP247672	Borrowdale Brooke	1800000	478,692.64	1,321,307.36
Closest prediction	SBY247456	Zimre Park	110000	109,054.83	945.17

For the largest-error case, the strongest local sensitivity is Harare North area membership (+0.5767 on the local log-price perturbation scale), followed by stand size (+0.2456). For the closest prediction, stand size is the dominant negative sensitivity (−0.4470), followed by visual kitchen and completion scores. These are perturbation-based local sensitivities, not SHAP values and not causal effects.

6.5 Fairness and proxy-discrimination risk

Suburb and broader area are legitimate valuation variables because location is strongly associated with observed market prices, but they can also proxy for socio-economic and demographic characteristics. For a regulator or model-risk committee, the appropriate defence is empirical and procedural: demonstrate that location materially improves out-of-suburb predictive performance; document the effect after controlling for observable property characteristics; monitor residuals and approval outcomes

across geographic groups; and avoid interpreting a location coefficient as a causal or demographic effect.

The present notebook provides evidence that suburb remains statistically associated with price after controlling for floor size and bedrooms and that removing location worsens test log-RMSE. However, this alone does not establish fairness. A production AVM should add richer property characteristics, spatial smoothing or coordinates, repeated out-of-area validation and explicit disparate-impact monitoring where legally and ethically appropriate.

7. Section F — Applied Case Study

7.1 Case specification and point estimates

The two synthetic properties are both specified as 300 m² stands with four bedrooms and two bathrooms, complete structural shells but unfinished internal/external paint, partial ceilings, unfitted kitchens, bare/screeded floors, unpaved and unfenced yards, municipal water, ZESA and no solar. House A is in Madokero, Harare West with title deeds. House B is in Mabvazuva, Ruwa with developer cession/title pending. Because floor size is not provided, the modelling pipeline imputes the training median. Because no case-study photographs are supplied, the visual scores are set to training medians.

Property	Location	Tenure	Point estimate (USD)	90% lower (USD)	90% upper (USD)
House A	Madokero, Harare West	Title deeds	95,349.07	17,578.78	517,182.89
House B	Mabvazuva, Ruwa	Developer cession/title pending	94,706.26	17,460.27	513,696.23

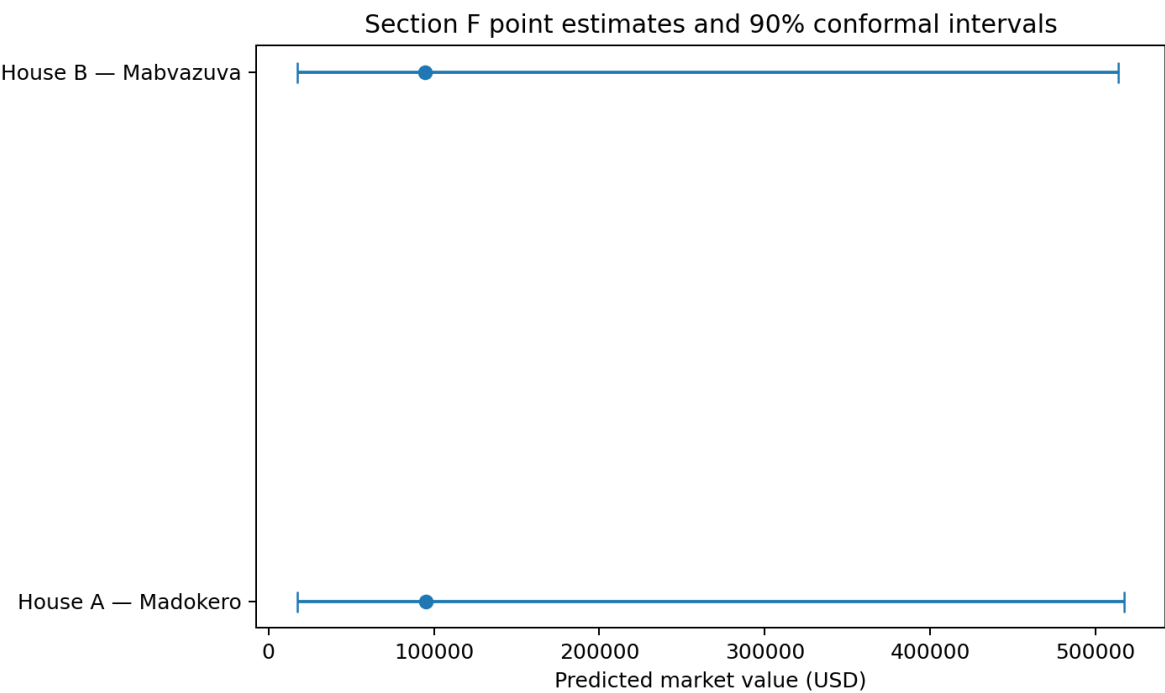


Figure 9. Case-study point estimates and 90% conformal prediction intervals.

7.2 Location and title decomposition

Quantity	Model value/effect (USD)
House A: Madokero + title	95,349.07
House B: Mabvazuva + cession	94,706.26
Mabvazuva + title counterfactual	95,156.98
Madokero + cession counterfactual	95,536.69
Location change, title held constant	-192.09
Title-to-cession effect at Madokero	-187.62
Title-to-cession effect at Mabvazuva	-187.24

The model attributes only a very small difference to the Madokero/Mabvazuva location change and to title-versus-cession status for these particular synthetic inputs. This should not be interpreted as a general market discount. It is a model-specific local sensitivity produced under median-imputed floor size and median visual scores.

7.3 Extrapolation and trust

Property	5-NN distance	Training-distance percentile (%)	Interpretation
House A — Madokero	2.3233	27.08	Within dense/common training region
House B — Mabvazuva	2.6066	70.83	Moderately sparse

The training 5-nearest-neighbour distance distribution has median 2.4829, 75th percentile 2.6327, 90th percentile 2.8096 and 95th percentile 4.0466. House A lies at the 27th percentile and is therefore not an extreme extrapolation. House B lies at about the 71st percentile and is moderately sparse but still below the 75th-percentile threshold. Four of House B's five nearest neighbours are actual Mabvazuva listings, which supports local relevance. Trust should nevertheless remain limited because the conformal intervals are very wide and the case-study finish state is represented indirectly.

7.4 One-page actuarial memo

To: Mortgage Risk Unit and Underwriting Department

Subject: AVM assessment of partially finished houses in Madokero and Mabvazuva

The multimodal AVM produces indicative market-value estimates of approximately \$95,349 for House A and \$94,706 for House B. The 90% conformal intervals are approximately \$17,579–\$517,183 and \$17,460–\$513,696 respectively. These intervals are too wide for the point estimates to be treated as stand-alone collateral valuations.

Location loading is defensible only to the extent that it reflects repeatable market evidence and survives controls for observable property characteristics. In this model, suburb is strongly associated with log price and removing location worsens out-of-sample log-RMSE. Nevertheless, geographic variables may proxy for omitted socio-economic characteristics, so the loading requires monitoring, transparent documentation and periodic fairness review.

For mortgage LTV, the AVM should be used as a screening or challenger tool rather than as the sole valuation for an unfinished property. A professional inspection remains important because finish quality, completion costs and title documentation materially affect recoverable collateral value. For insurance sums insured, market value is not the correct basis: a separate reconstruction-cost approach should be used, particularly for partially finished structures.

A concrete methodological improvement is to expand the training sample with more unfinished and partially finished properties and fit a hierarchical/spatial model in which condition effects can vary by locality. This would reduce reliance on broad suburb labels and should improve estimates for atypical properties.

8. Conclusions and Limitations

The assignment demonstrates that multimodal information can improve residential property prediction beyond a structured actuarial benchmark. Location and size remain dominant, but description text and image-derived finish scores add measurable predictive information. The suburb-disjoint split is a useful stress test because it evaluates performance on locations not seen during fitting.

The main limitations are the modest 150-listing multimodal cohort, sparse explicit condition phrases, the use of advertised rather than transacted prices, wide prediction intervals, imputation of missing case-study floor size, and median visual features for the synthetic houses. In addition, the scraping date remains unspecified in the notebook and should be completed before submission. Future work should enlarge the multimodal sample, incorporate geospatial coordinates and transaction data where available, strengthen uncertainty calibration and perform repeated spatial cross-validation.

9. Reproducibility and AI-use Disclosure

The submitted Jupyter notebook contains the scraping, cleaning, feature engineering, modelling, diagnostics and case-study workflow. The report should be submitted together with the notebook/code and the cleaned/model-ready data files required by the assignment brief.

AI-use disclosure: Generative AI was used to assist with explanation, report structuring and presentation of results. The numerical results, tables and model outputs reported here were taken from the student's executed Jupyter notebook. The student remains responsible for verifying the analysis, code, interpretation and final submitted wording in accordance with the University's academic-integrity requirements.

Appendix A — Key Model Configuration

Component	Configuration
Target	log(advertised USD price)
Structured numeric	bedrooms, bathrooms, floor_size_m2, stand_size_m2; median imputation + scaling
Categorical	suburb, area; most-frequent imputation + one-hot encoding
Text	TF-IDF 1–2 grams, min_df=3, max_df=0.90, sublinear TF; 30 SVD components
Visual	8 CLIP condition/finish scores
Final RF	500 trees; max_depth=5; max_features=0.7; min_samples_leaf=1; random_state=42

Appendix B — Notes for Final Submission Check

Before submitting, replace the missing scraping-date placeholder with the actual date from the scraping session. Confirm that the raw/cleaned data files and README are included with the code package. The report deliberately does not invent a ZiG exchange-rate conversion because the notebook documents a USD-only modelling strategy.